Program 4: Working with Selenium (PractiseAutomation.com)

Step 1: create a maven project:
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp02 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false

```
harith@harith-VirtualBox:~/devops/test$ mvn archetype:generate \
-DgroupId=com.example \
-DartifactId=MyMavenSeleniumApp02 \
-DarchetypeArtifactId=maven-archetype-quickstart \
-DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 10 MB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/harith/devops/test
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp02
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/harith/devops/test/MyMavenSeleniumApp02
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.656 s
[INFO] Finished at: 2026-03-22T15:06:24+05:30
[INFO] ------------------------------------------------------------------------
```

Step 2 : edit pom.xml file

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
  <modelVersion>4.0.0</modelVersion>
  <groupId>com.example</groupId>
  <artifactId>MyMavenPracticeAutomationApp</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>MyMavenPracticeAutomationApp</name>
  <url>http://maven.apache.org</url>


  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>


    <!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
    <groupId>org.seleniumhq.selenium</groupId>
    <artifactId>selenium-java</artifactId>
    <version>4.29.0</version>
</dependency>
  </dependencies>
```

```
  <!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.7</source>
<target>1.7</target>
</configuration>
</plugin>
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
            <groupId>org.apache.maven.plugins</groupId>

            <artifactId>maven-jar-plugin</artifactId>
            <version>3.0.1</version>
           <configuration>
                <archive>
                    <manifestEntries>
                        <Main-Class>com.example.App</Main-Class>
                    </manifestEntries>
                </archive>
            </configuration>
        </plugin>


</plugins>
</build>
</project>
```

Step 3: Edit the App.java

```
Open ~ App.java
~/devops/test/MyMavenSeleniumApp02/src/main/java/com/example

package com.example;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.chrome.ChromeDriver;

import java.time.Duration;

public class App {

    public static void main(String[] args) {

        WebDriver driver = new ChromeDriver();



        driver.get("https://practicetestautomation.com/practice-test-login/");

        driver.findElement(By.id("username")).sendKeys("student");
        driver.findElement(By.id("password")).sendKeys("Password123");
        driver.findElement(By.id("submit")).click();

        String heading = driver.findElement(By.tagName("h1")).getText();
        System.out.println("Page Heading: " + heading);

    }
}
```

Step 4:build and run the maven application

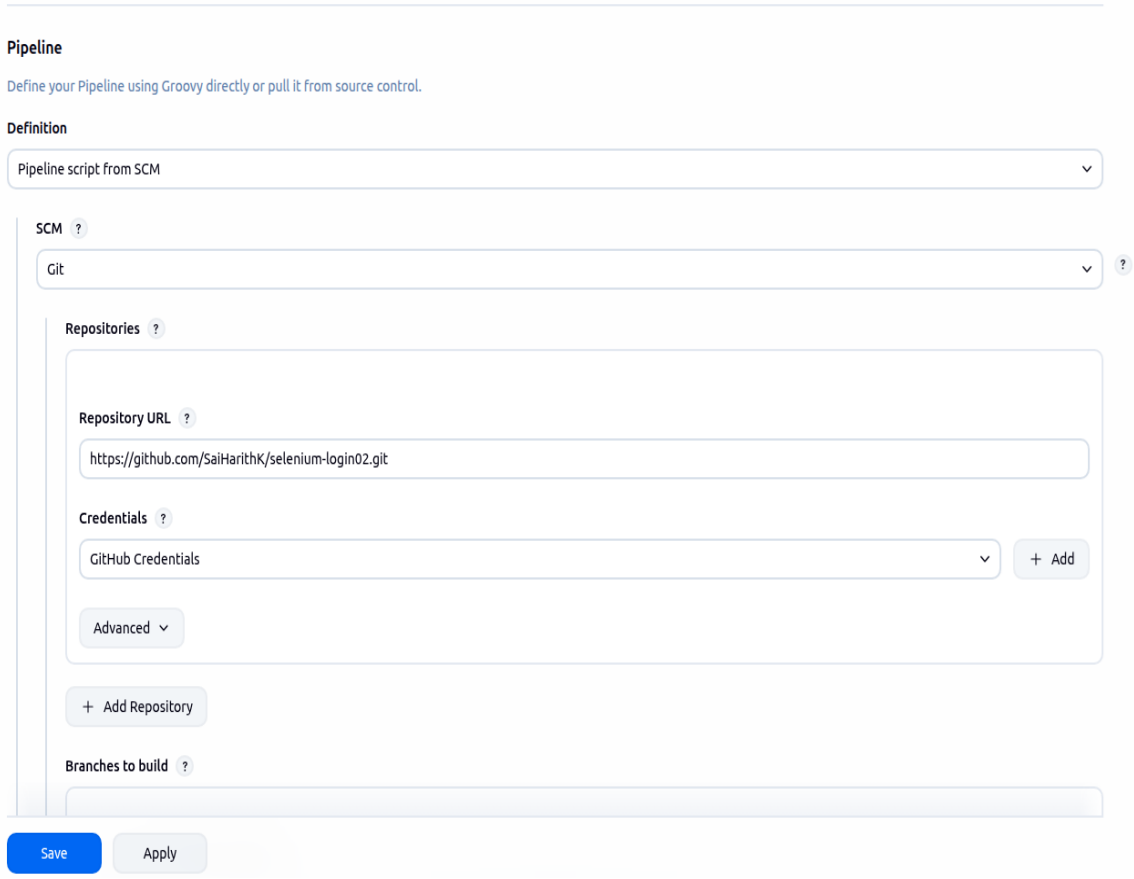
mvn clean install
mvn exec:java -Dexec.mainClass="com.example.App"

```
-------------------------------------------------------
 T E S T S
-------------------------------------------------------
Running com.example.AppTest
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.035 sec

Results :

Tests run: 1, Failures: 0, Errors: 0, Skipped: 0

[INFO]
[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ MyMavenFirefoxApp ---
[INFO] Building jar: /home/harith/devops/test/MyMavenSeleniumApp02/target/MyMavenFirefoxApp-1.0-SNAPSHOT.jar
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MyMavenFirefoxApp ---
[INFO] Installing /home/harith/devops/test/MyMavenSeleniumApp02/target/MyMavenFirefoxApp-1.0-SNAPSHOT.jar to /
le/MyMavenFirefoxApp/1.0-SNAPSHOT/MyMavenFirefoxApp-1.0-SNAPSHOT.jar
[INFO] Installing /home/harith/devops/test/MyMavenSeleniumApp02/pom.xml to /home/harith/.m2/repository/com/exa
MyMavenFirefoxApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.110 s
[INFO] Finished at: 2026-03-22T15:17:12+05:30
[INFO] ------------------------------------------------------------------------
harith@harith-VirtualBox:~/devops/test/MyMavenSeleniumApp02$
```

Step 5: Pipeline

-login to jenkins account

-create a new item and configure the following pipeline settings

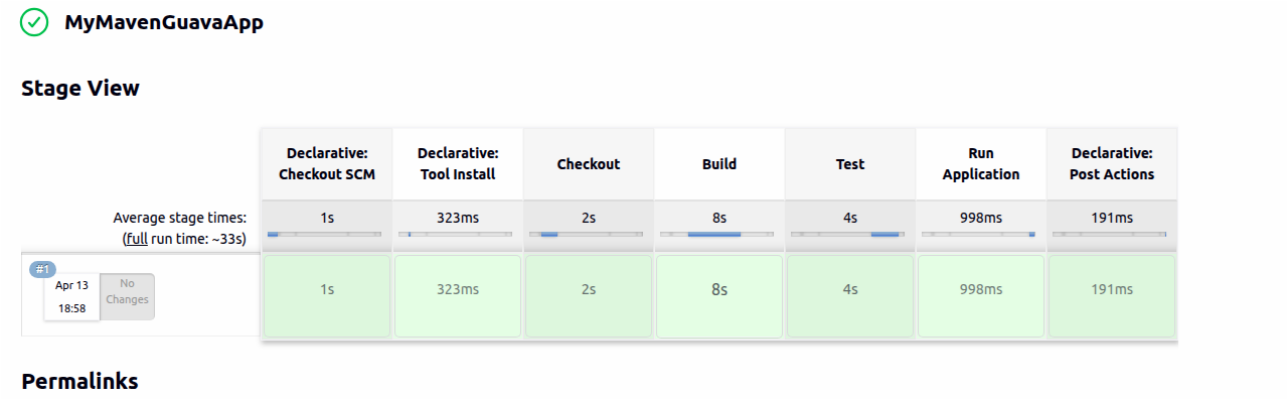


Step 6: Edit Jenkins file

-Create a Jenkinsfile and add the following code:

```
pipeline {
    agent any  // Use any available agent


    tools {
        maven 'Maven'  // Ensure this matches the name configured in Jenkins
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/SaiHarithK/selenium-login02.git'
            }
        }
        stage('Build') {
            steps {
                sh 'mvn clean package'  // Run Maven build
            }
        }
        stage('Test') {
            steps {
                sh 'mvn test'  // Run unit tests
            }
        }
stage('Check Target Folder') {
    steps {
        sh 'ls -l target'
    }
}
       stage('Run Application') {
    steps {
        sh 'java -jar target/selenium-login-1.0-SNAPSHOT.jar'
    }
```

```
}}
    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

Step 7: Build

Click on build now and check if its successfull